INCH-POUND
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SUPERSEDING
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6 December 1985

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, FILM, NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, AND SPACE LEVEL, STYLE RNR50

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for style RNR50 nonestablished reliability, established reliability, and space level, film, fixed resistors. 1/

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-55182 - Resistors, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, General Specification for.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications - Electronics Command and Fort Monmouth, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
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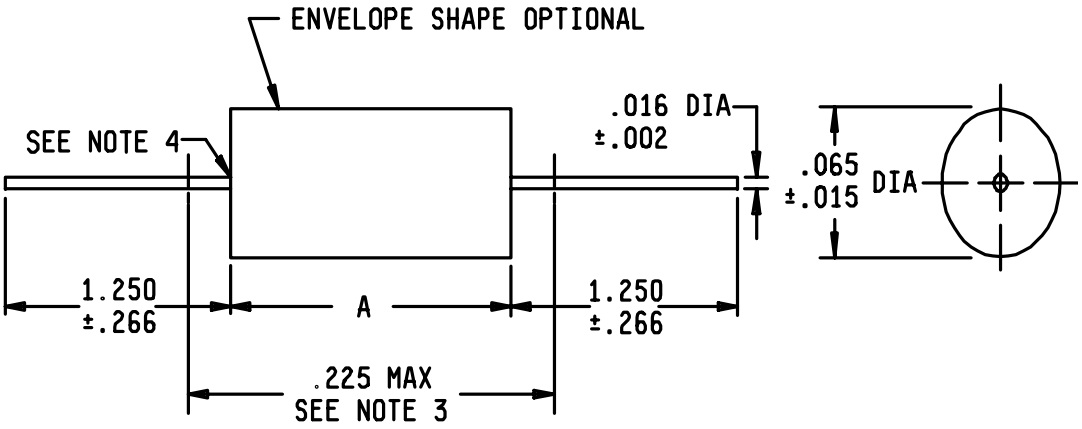

Characteristic	Dimension A ±0.020
C, E	0.180
H, J, K	0.150

Inches	mm	Inches	mm
.002	0.05	.150	3.81
.015	0.38	.180	4.57
.016	0.41	.225	5.72
.020	0.51	.266	6.76
.065	1.65	1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The maximum length is "clean lead" to "clean lead".
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.

FIGURE 1. Style RNR50 resistors. 2/

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-55182G.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1.

3.2.1 Characteristic. Style RNR55 is available in characteristic C, characteristic E, characteristic H, characteristic J, and characteristic K.

3.2.2 Terminal. Terminal types RNR, RNC, and RNN are available in all characteristics. Terminal RNR is inactive for design with characteristic H, characteristic J, and characteristic K.

2/ Reference footnote 1.

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3.3 Power rating. The power rating shall be 0.05 watt (see 6.2).

3.4 Voltage rating. The continuous working voltage shall not exceed 200 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as specified in table I.

TABLE I. Minimum and maximum resistance values.

Minimum resistance		Maximum resistance	
Characteristic	Ohms	Characteristic	Megohms
H	10.0	H	0.796
J	10.0	J	0.796
K	10.0	K	0.796
C	10.0	C	0.1
E	10.0	E	0.1

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.005 percent per volt.

3.7 Mechanical shear. The mechanical shear test is applicable to this specification.

3.8 Marking. Due to size limitations this style resistor shall be marked with the following minimum information:

042K - Date code, characteristic.
49R9 - Coded resistance value.
FMCJ - Tolerance, product level, lead material, JAN marking.
* - Manufacturer's code (see qualified products list for manufacturer's code symbols).

4. VERIFICATION

4.1 Verification. Verification shall be in accordance with MIL-PRF-55182.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-55182 are applicable to this specification.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. Packaging requirements (see 5.1).

6.3 Power rating at 70°C. The power rating at 70°C for 2,000 hours life test has been established at 0.1 watt, maximum voltage 200 volts. It should be noted that the failure rate level is established at the single condition of 125°C, 0.05 watt and 10,000 hour life test duration.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

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